CRISP-DM Template for SDG Dashboard Design

This template will guide you through the CRISP-DM process while designing a dashboard focusing on one of the Sustainable Development Goals (SDG) indicators. Each section corresponds to a step in the CRISP-DM cycle. Provide detailed explanations and documentation of your work at each stage.

1. Business Understanding

In this section, define the objectives of your dashboard, which is the data-driven research question. Identify the specific SDG indicator you will focus on and explain the importance of this indicator to the broader SDG goals. Clearly state what you aim to achieve with the dashboard.

Key Questions to Address:

- - What is the purpose of the dashboard?
  I come from Iran whereas I showed in my dashboard I experienced big inflation and I saw the differences of overall health in 5 years and I decide to make a research on it to see does it have same result in different countries or not and I was trying to show to other students to what extend it can effect on society.
- - Which SDG indicator is you focusing on?
  I chose 2 SDG indicators: No Poverty and Zero Hunger
  because as I showed in my dashboard inflation caused poverty in many countries and poverty was the reason of getting malnutrition and Zero Hunger. If we cannot manage worldwide inflation, we will lose many people from hunger.
- - What insights do you hope to provide with this dashboard?
  I hope to manage worldwide inflation and care more about the people who are dying from hunger or malnutrition. Or set a equality for nutrition all over the world and all countries provide a good nutrition for their people. Having a institute to manage it.

2. Data Understanding

Describe the data you are using to build the dashboard. Explain the sources of the data, the variables included, and any initial findings from exploring the data. Highlight any data quality issues or unusual patterns.

Key Questions to Address:

- - What are the sources of your data?
  To have accurate and correct results I collect my data from different sources. My data was collected from Statista, Our world in data, WHO (world health organization)

And my sources are about the worldwide inflation and average eon inflation in different continents, food index in 10 years, the reason of increasing inflation in 2020, number of total deaths around the world by malnutrition, average of people who cannot afford healthy diet in 10 years and life expectancy of all countries in 10 years.

- - What variables are included in the dataset?

In my dashboard I was trying to understand each year how much we had a inflation and to what extend it related to the number of death and number of people dying from malnutrition. Well, years and number of people who die are my variable.

- - Are there any data quality issues?
  Well, no there were not any data quality issues just because it was from different sources it was hard to clean it and make a relation with other data.
- - What initial insights did you gain?
  My initial insight that I got from my dashboard, honestly, I believe all the SDG indicators related to each other's you cannot fix only one of them. You must work to all the indicators to change something in features.
  But from my dashboard I saw that there is a lot of inequality all over the world.

3. Data Preparation

Document the steps you took to clean, transform, and prepare the data for analysis. This may include removing duplicates, handling missing values, and creating new variables that will be useful for the dashboard.

Key Questions to Address:

- - What data cleaning processes did you perform?
  Basically, I did clean all my columns and removed errors from all of them to get a accurate error. I removed some columns that were not necessary to be there. To can compare my data I did normalized to some my data. Use the first row as a header to have clear data. Sort them based on year because my data was mostly compared based on year. Remove some rows that have a different value.
  Checked the dictation of some of my data to have same value in different table.
  Make relation with my different table.
- - How did you handle missing values?
  On the top of the page in the view part I could see which of my data is empty or how much from what data I have so on that part I can understand what I should remove or should I do remove duplicate or just change the manually.
- - Did you create any new variables or features?
  Yes, as I said, I normalized some of my data can compared it because one of them it was about 1 to 100 and the other one is about a million so I make both of them about a hundred to can compare it.

4. Modeling

Explain any models or algorithms you used to analyze the data. If your dashboard includes predictive elements, describe the models and how they were trained. If not, explain how you structured the data to create meaningful visualizations.

Key Questions to Address:

- - What models or algorithms did you use?

To investigate the links between variables like inflation rates, food costs, and health outcomes, the analysis mostly uses statistical and descriptive models, including correlation analysis. To clearly display trends and distributions, basic aggregation methods are utilized, such as computing averages and median values over years and geographies.

- - How did you ensure the model's accuracy?

The dashboard compares data from different years, geographies, and variables in order to cross-validate anticipated patterns and ensure accuracy. This involves verifying that undernourishment rates and food price indices logically connect with inflation patterns. The dashboard's computations were checked to make sure they provide accurate insights and match anticipated health and economic trends.

- - How did you structure the data for visualization?

An understandable narrative is made possible by the data's year-, region-, and socioeconomic indicator-based structuring across visualizations. To examine factors like inflation and health disparities across continents, the data is made visually intuitive and accessible using pie charts for distribution summaries, stacked column charts for regional comparisons, and line charts for time-series data.

5. Evaluation

Evaluate the results of your modeling or visualization efforts. Discuss whether the dashboard meets the original objectives set out in the business understanding phase. Identify any limitations and how they might impact the dashboard's effectiveness.

Key Questions to Address:

- - Does the dashboard meet the business objectives?

Indeed, by clearly demonstrating how inflation affects poverty, health, and the availability of a nutritious diet (in line with SDG 1: No Poverty and SDG 2: Zero Hunger), the dashboard satisfies the initial goals from the business understanding phase.

- - What are the limitations of your analysis?

It was challenging to find an accurate answer because my research question was new and I had to gather data from various sources, but I did my best to identify connections between them so that I could create a useful dashboard.

- - How could the dashboard be improved?

Adding dynamic filters that let users examine data by demographic characteristics (such age or income level) could improve the dashboard. The research would be further enhanced by adding real-time data and more thorough country-level insights.

6. Deployment

Outline the steps required to deploy the dashboard for use. This may include publishing it to a platform like Power BI, sharing it with stakeholders, and ensuring it is regularly updated with new data.

Key Questions to Address:

- - How will you deploy the dashboard?

Stakeholders will be able to easily view the dashboard thanks to Power BI's sharing capabilities. Because Power BI is cloud-based, users can view the dashboard from any device, making it convenient for analysis and decision-making.

- - Who is the target audience for the dashboard?

Policymakers, economic analysts, and health groups that prioritize poverty alleviation and socioeconomic development are the target consumers. These observations can help make well-informed policy changes in regions where food shortages and inflation are major issues.

- - How will you ensure the dashboard stays up to date?

Power BI's regular data refreshes will keep the dashboard up to date by enabling automated updates whenever new data becomes available, guaranteeing stakeholders' continued relevance.

7. Future Research

List down the steps that might address the shortcomings in your research. These could be availability of data, outdated research data.

Key Questions to Address:

- Are there emerging techniques or tools that could provide deeper insights?- Who is the target audience for the dashboard?

Indeed, new methods like predictive analytics and machine learning have the potential to greatly enhance the dashboard's depth. Predictive models, for example, could predict the potential effects of future inflation on food prices and health outcomes, enabling stakeholders to plan ahead and take proactive measures. A useful dimension could be added to the dashboard by using tools like social media sentiment analysis, which could offer real-time insights into public perceptions and prompt responses to economic developments.

- How might the research impact different stakeholders or communities?

Many stakeholders may be impacted by the dashboard's research insights. While food security-focused NGOs could direct resources to the most impacted communities, policymakers could use the findings to modify social policies to lessen the impact of inflation on low-income groups. Furthermore, in regions where food affordability is impacted by inflation, healthcare organizations could predict service demand, assisting them in efficiently allocating resources to promote community health. This template is designed to help you systematically document your work at each stage of the CRISP-DM process, ensuring a comprehensive and well-structured approach to designing your SDG dashboard.